**Advantages of Lithium Battery**

The are many advantages of using lithium-ion battery.

**i) High Energy Density**

The main advantage of lithium ion battery is high energy density. In mobile phones, it needs to operate for a long time between charges while still charging more power, there is always a need to batteries, with a much higher energy density. It is a distinct advantage.

**ii) Self Discharge**

One issue with batteries is that they loose their charges overtime. The main advantage is that the rate of self-discharge is very low than that of the other batteries.

**iii) No Requirement For Priming**

In this, lithium ion battery does not need to be primed, but the other batteries require priming.

**iv) Low Maintenance**

Lithium ion battery does not require any maintenance to ensure the performance.

**Disadvantages**

The disadvantages of lithium ion battery are as follows:

**i) Protection Required**

Lithium ion cells require protection from being overcharged and discharged too much. In addition, they need to have the current maintained within safe limits. Accordingly, lithium ion battery disadvantage is that they require protection to ensure that is it kept within the safe operating limits.

**ii) Ageing**

Another disadvantage of this battery is ageing. The battery is dependent upon the number of charge and discharge cycles that the battery has undergone. Lithium ion battery should be kept in a cool storage area, that will increase the life of battery.

**iii) Transportation**

Lithium ion battery applications are restricted on their transportation, especially by air. These batteries require care and protection while on transportation.

**iv) Cost**

The cost of lithium ion battery is high compared with other types of batteries.

## 4.6 SEVEN FEATURES ABOUT THE DISPARITY BETWEEN LEAD ACID AND LITHIUM ION BATTERIES

**1. Weight**

Lithium ion batteries are one third the weight of lead acid batteries

**2. Efficiency**

Lithium-ion batteries are of nearly 100% efficiency both charge and discharge, allowing the same ampere hours both in and out. But lead acid cell battery is 85% efficiency.

**3. Discharge**

Lithium-ion batteries are discharged 100%, but lead acid batteries discharge less than 80%.

**4. Life Cycle**

Life cycle of the lithium-ion battery is 400–1200 cycles, whereas lead acid battery life cycle is 400 to 500 cycles

**5. Voltage**

Lithium-ion batteries maintain their voltage throughout the entire discharge cycle. This allows greater and longer lasting efficiency of electrical components. Lead



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